

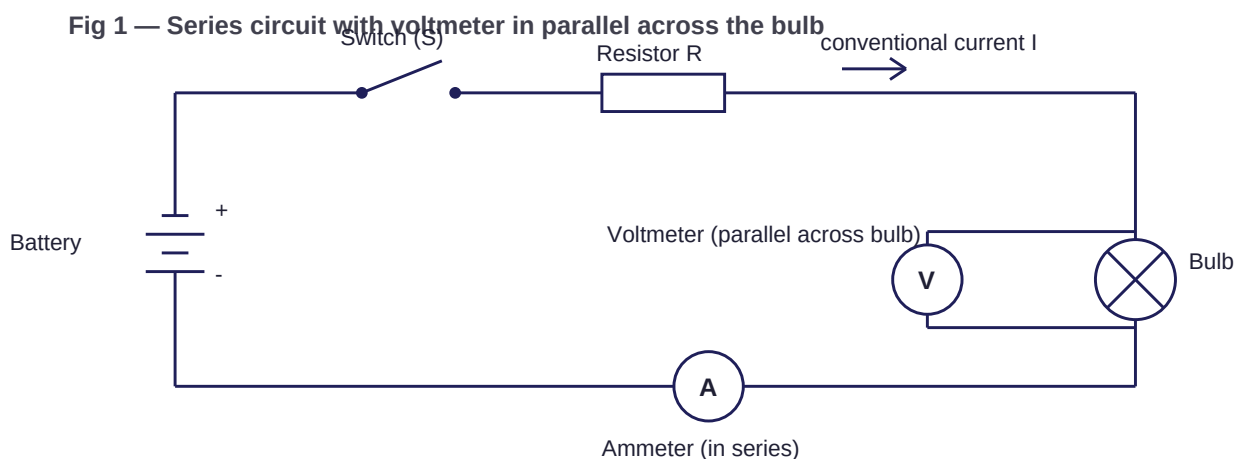
Question 1 — Science (5 marks)

An electric circuit is a closed conducting path through which current can flow. The battery provides the potential difference that pushes the current around the circuit. The switch is used to open or close this path. When the switch is closed the path is complete and current flows through the resistor and the bulb; when it is open the path is broken and no current flows at all.

In my diagram the battery, switch, resistor, bulb and ammeter are all joined in one closed loop, that is, in series. The ammeter has to be in series because the current it measures is the same current that passes through the components. The voltmeter is connected in parallel across the bulb, because it measures the potential difference between the two ends of the bulb and not the current through it.

Conventional current is taken to flow from the positive terminal of the battery, through the switch, resistor, bulb and ammeter, and back into the negative terminal. I have marked this direction with an arrow on the diagram.

The current depends on the voltage and the resistance together. By Ohm's law $V = IR$, so $I = V/R$. If the battery voltage is kept the same and the resistance is increased, the current becomes smaller. If the resistance is reduced, a larger current flows. The bulb would therefore glow more dimly if I increased the resistance.

Fig 1 — Series circuit with voltmeter in parallel across the bulb

Question 2 — English (5 marks)

Technology has changed how students learn because information that was once hard to reach is now available in seconds. A student can use digital libraries, videos and interactive tools to understand a concept that did not make sense in class. This makes learning flexible, because each student can go at their own pace. For example, when I could not follow a chemistry topic in class, I watched three different explanations online until one of them finally made sense to me.

On the other hand, easy access can create dependence. A student who searches for the answer to every hard question finishes the homework but never builds the habit of working a problem out. There is also the risk of accepting wrong information just because it appeared on a website that looked trustworthy. Somebody who only copies is really practising searching, not thinking.

In my opinion technology does not automatically make students better or worse learners; the effect depends entirely on how it is used. It helps most when it is used to understand a concept, compare explanations and check your own work, and it harms when it replaces the effort of thinking. So I would say technology should be treated as a tool that supports thinking rather than a substitute for it.

Question 3 — Economics (5 marks)

I have plotted quantity on the horizontal axis and price on the vertical axis. The demand curve slopes downward from left to right because consumers buy more when the price falls. The supply curve slopes upward from left to right because producers are willing to sell more when the price rises.

The two curves cross at a price of Rs 30 and a quantity of 60 units. This is the market equilibrium, because at this price quantity demanded is exactly equal to quantity supplied and there is no pressure for the price to change.

If the market price is below Rs 30, say Rs 20, then 80 units are demanded but only 40 are supplied. This is a shortage: buyers want more than sellers will supply, so the price is pushed upward. If the price is above Rs 30, say Rs 40, then 80 units are supplied but only 40 are demanded. This is a surplus, and unsold stock pushes the price back downward. Either way the market moves back towards the equilibrium.

If the cost of production rises, producing each unit is less profitable at the old price, so producers supply less at every price. The whole supply curve therefore shifts to the left (upward), shown as S1 on my graph. Assuming demand does not change, the new equilibrium is at a higher price and a lower quantity than before.

Fig 2 — Demand and supply, with the supply curve after a cost increase

